

Appendix to DNB Analysis “Healthy markets, healthy growth: Dynamics, market power and misallocation in the Dutch business sector”

Guus Brouwer*

Jasper de Winter[†]

9 juni 2026

*Economics and Research Division, De Nederlandsche Bank, g.s.p.brouwer@dnb.nl.

[†]Economics and Research Division, De Nederlandsche Bank, j.m.de.winter@dnb.nl.

A Data and Estimation of Markups, Markdowns, and Misallocation

In this section, we describe the CBS microdata used and the estimation of markups, markdowns, and misallocation. The microdata are accessible exclusively within the secure CBS microdata environment (see: <https://www.cbs.nl/microdata>).

A.1 Microdata Used

All datasets used are derived from confidential microdata of Statistics Netherlands (CBS), which are made available to accredited research institutions through the secure CBS research environment. The dataset covers the period 2007–2023 and, in line with the main text of the DNB Analysis, focuses on the delineated market sector.

The Statistics on the Finances of Non-Financial Corporations (NFO) form the core of our dataset. The NFO is a CBS-compiled microdata source with detailed annual information on the balance sheet and profit-and-loss accounts of virtually all non-financial corporations with legal personality (private and public limited companies) in the Netherlands. The data are based on a combination of primary surveys of large firms (SFGO) and administrative data from corporate income tax filings of smaller firms (SFKO). The NFO allows us to consistently measure firms’ financial variables, including turnover, value added, capital, and intermediate inputs.

Information on labor input is not available in the NFO and is therefore obtained from the General Business Register (ABR) of CBS. The ABR contains core information for all active firms, such as industry classification at a detailed SBI-4D level and the number of workers. Using the enterprise group identification number (`ond-id`), which is consistent across the NFO and ABR, we link financial data from the NFO to labor data from the ABR at the enterprise group level. This combination enables us to analyze markups, markdowns, and misallocation of labor and capital for a large and consistent population of firms. In line with [Adema et al. \(2015\)](#), we exclude firms with fewer than two workers. Per year, our sample contains data on approximately 300,000 enterprise groups.

A.2 Markups

The markup is defined as the ratio of the output price to marginal cost ([De Loecker en Warzynski, 2012](#)). Estimating markups is not only relevant as an indicator of product market power, but also as a building block for the estimation of markdowns. Without the

wedge between prices and marginal costs, one cannot isolate the markdown attributable to labor market power alone, since part of the gap between the marginal revenue product of labor and the actual wage would be confounded by output market power. As such, the estimation of markups directly links to our estimation of markdowns (see Section A.3) and misallocation (see Section A.4).

To compute the markup following the method of [De Loecker en Warzynski \(2012\)](#), we assume the following production function:

$$Y_{is,t} = Y_{is,t}(M_{is,t}, L_{is,t}, K_{is,t}; \omega_{is,t}), \quad (1)$$

where $Y_{is,t}$ denotes output, $L_{is,t}$ labor, $K_{is,t}$ the sum of the volume of tangible and intangible fixed assets, $M_{is,t}$ the sum of the volume of raw materials, intermediate inputs, and other operating expenses, and $\omega_{is,t}$ firm-level productivity for firm i in sector s in year t .

The first-order condition for the variable/flexible input, assumed to be free of adjustment costs ($M_{is,t}$), is:

$$\mu_{is,t} = \frac{p_{is,t}^Y}{\lambda_{is,t}} = \theta_{is,t}^M * \frac{p_{is,t}^Y Y_{is,t}}{p_{is,t}^m M_{is,t}}. \quad (2)$$

Here, $\mu_{is,t}$ denotes the markup of firm i in sector s in year t , $\lambda_{is,t}$ marginal cost, $\theta_{is,t}^M$ the output elasticity of the flexible input, and $\frac{p_{is,t}^Y Y_{is,t}}{p_{is,t}^m M_{is,t}}$ the ratio of revenues to expenditures on that input.

Both $p_{is,t}^Y Y_{is,t}$ and $p_{is,t}^m M_{is,t}$ are directly observed by CBS, although prices and volumes are not separately observed. Empirically estimating the markup therefore hinges on estimating the output elasticity $\theta_{is,t}^M$. We do so in two ways. The first method—which we use in the main text of the DNB Analysis—is based on the Olley–Pakes production function approach and is described in Section A.2.1. In Section A.2.2, we describe the alternative markup estimation approach.

A.2.1 Estimating the Output Elasticity Following [Olley en Pakes \(1996\)](#)

In the DNB Analysis, markups are computed using a sector-specific Cobb–Douglas production function:

$$Y_{is,t} = \omega_0^{\beta_0} K_{is,t}^{\beta_k} L_{is,t}^{\beta_l} M_{is,t}^{\beta_m} e^{\varepsilon_{is,t}}. \quad (3)$$

Here, ω_0 denotes a sector-specific constant, and $\varepsilon_{is,t}$ is an unobservable firm-, sector-, and time-specific factor affecting a firm’s production volume (such as a productivity shock).

Taking natural logarithms yields the following log-linear specification:

$$y_{is,t} = \beta_0 + \beta_k k_{is,t} + \beta_l l_{is,t} + \beta_m m_{is,t} + \varepsilon_{is,t}. \quad (4)$$

This equation is estimated separately for the 25% largest and 75% smallest firms, measured by turnover, allowing for differences in production technologies between large and small firms. To obtain real variables, we use sector-specific deflators for turnover, investment, and intermediate inputs, in line with [van Heuvelen et al. \(2019\)](#).

Our primary interest lies in β_m in equation (4), as this parameter directly enters the firm-level markup calculation. While we distinguish between large and small firms, we assume that this elasticity is otherwise constant within sectors and over time.

Equation (4) could in principle be estimated using ordinary least squares (OLS), but this leads to endogeneity problems. Firms' choices of capital, labor, and materials typically depend on unobserved factors that also affect productivity. If producers have (partial) information about their productivity at the time input decisions are made, the regressors will be correlated with the error term. This violates the exogeneity assumption and results in upward-biased OLS estimates ([Akerberg et al., 2015](#)).

To make this explicit, the error term ($\varepsilon_{is,t}$) is decomposed into a predictable productivity component ($v_{is,t}$) and an unanticipated shock ($\xi_{is,t}$):

$$y_{is,t} = \beta_0 + \beta_k k_{is,t} + \beta_l l_{is,t} + \beta_m m_{is,t} + v_{is,t} + \xi_{is,t}, \quad (5)$$

where $v_{is,t}$ represents productivity that is (partially) observed by the producer. Crucially, input choices depend on $v_{is,t}$, rendering OLS inconsistent.

[Olley en Pakes \(1996\)](#) propose a solution that exploits differences in input flexibility and information contained in investment decisions. Materials are assumed to be fully flexible within period t , whereas capital and labor result from decisions made in period $t - 1$. Productivity is assumed to evolve according to a first-order Markov process, and investment is assumed to contain information about current productivity. Under these assumptions, the investment decision can be written as

$$i_{is,t} = f_t(k_{is,t}, l_{is,t}, v_{is,t}), \quad (6)$$

which can be inverted to yield

$$v_{is,t} = f_t^{-1}(k_{is,t}, i_{is,t}, l_{is,t}). \quad (7)$$

Substituting this expression into equation (5) yields:

$$y_{is,t} = \beta_0 + \beta_k k_{is,t} + \beta_l l_{is,t} + \beta_m m_{is,t} + f_t^{-1}(k_{is,t}, l_{is,t}, l_{is,t}) + \xi_{is,t}. \quad (8)$$

Because the functional form of $f_t^{-1}(\cdot)$ is unknown, it is approximated by a flexible polynomial in capital, investment, and labor. Although this induces collinearity with the quasi-fixed inputs, the output elasticity of materials β_m remains identifiable, because material inputs are not included in the polynomial. For the purpose of estimating the markup, a consistent estimate of β_m suffices; the remaining input elasticities can be identified in a subsequent step.

A drawback of the Olley–Pakes method is that it requires firm-level investment data. Observations with zero or negative investment are therefore dropped, which can reduce sample size and lead to selection bias.

A.2.2 Estimating the Output Elasticity Following [De Loecker et al. \(2026\)](#)

[De Loecker et al. \(2026\)](#) develop a method to estimate markups without explicitly estimating a industry specific production function. This is attractive because production function estimation is empirically challenging. Unobserved productivity shocks affect both output and input choices, leading to endogeneity. Moreover, the solution provided by Olley and Pakes requires additional information about investments, leading to fewer observations and may lead to selection bias. The proposed approach circumvents these issues by deriving markups from the optimal choice of flexible inputs, without requiring additional firm-level investment information.

Specifically, the output elasticity is estimated at the sector-year level as:

$$\theta_{s,t}^M = \text{median} \left(1.1 * \frac{p_{is,t}^m M_{is,t}}{p_{is,t}^C C_{is,t}} \right), \quad (9)$$

where $p_{is,t}^C C_{is,t}$ denotes total production costs of firm i in year t , and $p_{is,t}^m M_{is,t}$ expenditures on the flexible input. The factor 1.1 reflects the assumption of modest economies of scale of this magnitude. By estimating the elasticity at the sector-year level, it is allowed to vary across sectors and over time.

In contrast to the method of [Olley en Pakes \(1996\)](#), which assumes constant output elasticities within sectors, this approach allows technological characteristics to evolve over time. Given the estimated output elasticity, the firm-level markup follows directly from

the ratio of revenues to expenditures on the flexible input:

$$\mu_{is,t} = \theta_{s,t}^M * \frac{p_{is,t}^Y Y_{is,t}}{p_{is,t}^m M_{is,t}}. \quad (10)$$

A.2.3 Decomposition

To shed light on the sources of changes in markups at the macro level, we employ a firm-level shift-share decomposition as proposed by [De Loecker et al. \(2020\)](#). This decomposition allows changes in the average markup to be broken down into contributions from within-firm adjustments, shifts in market shares across firms, and entry and exit dynamics. As such, it aligns with the broader literature that interprets macro-level developments in market power as the outcome of micro-level reallocation processes.

The change in the revenue-weighted average markup in the economy can be expressed as:

$$\begin{aligned} \Delta \bar{\mu}_t = & \underbrace{\sum_i (\mu_{i,t} - \mu_{i,t-1}) a_{i,t-1}}_{\text{within effect}} \\ & + \underbrace{\sum_i (a_{i,t} - a_{i,t-1}) (\mu_{i,t-1} - \bar{\mu}_{t-1})}_{\text{market share}} + \underbrace{\sum_i (a_{i,t} - a_{i,t-1}) (\mu_{i,t} - \mu_{i,t-1})}_{\text{cross term}} \\ & \underbrace{\hspace{10em}}_{\text{between effect}} \\ & + \underbrace{\sum_{i \in \text{entry}} (\mu_{i,t} - \bar{\mu}_{t-1}) a_{i,t} - \sum_{i \in \text{exit}} (\mu_{i,t-1} - \bar{\mu}_{t-1}) a_{i,t-1}}_{\text{net entry effect}}. \end{aligned} \quad (11)$$

Here, $\mu_{i,t}$ denotes the (unweighted) markup of firm i in period t , $\bar{\mu}_t$ the revenue-share-weighted average markup across the entire economy, and $a_{i,t}$ the revenue share of firm i at time t .

The first term, the *within effect*, captures changes in markups within incumbent firms, weighted by their initial market shares. This component reflects, for example, price adjustments or cost structure changes among incumbents, conditional on their existing market positions.

The *between effect* measures the extent to which changes in the average markup are driven by shifts in market shares across firms. This effect can be decomposed into a pure market share component—where firms with relatively high or low markups gain or lose market share—and a cross term measuring the extent to which changes in market shares coincide with changes in firm-level markups. This component aligns with the literature

linking market power to the reallocation of economic activity toward firms with higher markups.

Finally, the *net entry effect* captures the contribution of entering and exiting firms. Entry reduces (increases) the average markup when entrants have lower (higher) markups than the incumbent average, while exit has the opposite effect. This term therefore reflects the role of selection and creative destruction, as emphasized in the dynamic literature on market power and productivity.

A.3 Markdowns

Where the markup measures the ratio between price and marginal cost in the output market, the markdown measures the wedge between the marginal revenue product of labor and the actual compensation of labor in the labor market. This aligns with the misallocation framework, which focuses on differences in marginal revenue products across firms.

For the estimation of markdowns, we follow [Yeh et al. \(2022\)](#). They show that the markdown can be derived by combining the estimated markup with the output elasticity of labor. This requires estimating a production function using an appropriate flexible input other than labor. Under the assumption that labor is chosen statically and does not involve adjustment costs, the markdown is given by:

$$\nu_{is,t} = \frac{\theta_{s,t}^L}{\mu_{is,t}} * \frac{p_{is,t}^Y Y_{is,t}}{w_{is,t} L_{is,t}}. \quad (12)$$

Here, $\nu_{is,t}$ denotes the markdown of firm i in sector s and year t , $\mu_{is,t}$ the previously estimated markup, $\theta_{s,t}^L$ the output elasticity of labor, and $\frac{p_{is,t}^Y Y_{is,t}}{w_{is,t} L_{is,t}}$ the ratio of revenues to wage costs.

The intuition is that under monopsonistic buying power in the labor market, the marginal wage cost increases with the amount of labor hired. The first-order condition for labor can then be written as:

$$w'_{is,t}(L_{is,t})L_{is,t} + w_{is,t}(L_{is,t}) = \lambda_{is,t} \frac{\partial Y_{is,t}(\cdot)}{\partial L_{is,t}}, \quad (13)$$

where $\lambda_{is,t}$ represents the shadow value of production for firm i . As a result, the marginal revenue product of labor may exceed the actual wage rate. Rewriting this condition and using the definition of the markup yields equation (12).

In this analysis, the markdown $\nu_{is,t}$ is computed based on the markup estimate from

Section A.2. The output elasticity of labor $\theta_{s,t}^L$ is estimated using the production function approach of [Olley en Pakes \(1996\)](#), while the ratio of revenues to wage costs is directly observed in the data.

In practice, however, it is plausible that labor input is not fully static. Expanding or contracting the workforce is typically associated with search costs, administrative burdens, and firing costs. As a result, the wedge between marginal revenue and wage costs reflects not only employer market power, but also labor adjustment costs. To correct for this, we follow [Yeh et al. \(2022\)](#) and apply a correction to the initially estimated markdowns. In the presence of adjustment costs, the following holds:

$$\frac{R'_{is,t}(L_{is,t}^*)}{w_{is,t}(L_{is,t}^*)} = \nu_{is,t} + A_{is,t}(L_{is,t}^*, L_{is,t-1}), \quad (14)$$

where $A_{is,t}(\cdot)$ captures the contribution of adjustment costs. Assuming quadratic adjustment costs,

$$\Phi(L_{is,t}, L_{is,t-1}) = \frac{\gamma}{2} \left(\frac{L_{is,t} - L_{is,t-1}}{L_{is,t-1}} \right)^2, \quad (15)$$

with $\gamma \geq 0$, the uncorrected markdown $\hat{\nu}_{is,t}$ can be adjusted as follows:

$$\nu_{is,t} = \frac{\hat{\nu}_{is,t} - \gamma * (g_{is,t}^L(1 + g_{is,t}^L) - \beta g_{is,t+1}^L(1 + g_{is,t+1}^L)(1 + g_{is,t+1}^w))}{1 + \frac{\gamma}{2}(g_{is,t}^L)^2}. \quad (16)$$

We set the discount factor β equal to 1 and the adjustment cost parameter γ equal to 0.185, in line with [Yeh et al. \(2022\)](#). Growth in hours worked g^L and growth in total wage costs g^w are computed at the firm level using the available data.

A.4 Misallocation

To estimate the effect of misallocation of resources within industries on aggregate total factor productivity, we use the framework of [Hsieh en Klenow \(2009\)](#), which has recently been applied by, among others, [Gopinath et al. \(2017\)](#), [Gamberoni et al. \(2016\)](#), [Calligaris \(2015\)](#), and [Calligaris et al. \(2018\)](#). The Hsieh–Klenow model shows that frictions that distort the marginal revenue products of capital and labor reduce aggregate factor productivity.

In the Hsieh–Klenow model, aggregate output is described by a Cobb–Douglas production technology, while at the industry level output is a CES (constant elasticity of substitution) aggregate of differentiated products. For comparability with [Hsieh en Klenow \(2009\)](#) and [Gopinath et al. \(2017\)](#), we set the elasticity of substitution σ in the CES

function equal to 3.

Each individual firm produces its unique good according to a standard Cobb–Douglas production function:

$$Y_{is,t} = \omega_{is,t} K_{is,t}^{\alpha_s} L_{is,t}^{1-\alpha_s}. \quad (17)$$

We measure nominal firm-level value added, $p_{is,t}^Y Y_{is,t}$, as gross revenues minus the costs of materials used in production. Since we do not observe firm-level prices, we use two-digit industry deflators, following, for example, [Dias et al. \(2016\)](#), [Gorodnichenko et al. \(2018\)](#), and [Gopinath et al. \(2017\)](#). In practice, this implies that we approximate $Y_{is,t}$ as $p_{is,t}^Y Y_{is,t} / p_{s,t}^Y$, where $p_{s,t}^Y$ is the industry deflator.¹

We measure labor input $L_{is,t}$ as the firm’s wage bill, deflated using the same two-digit industry deflators applied to nominal output, i.e. as $W_{is,t} L_{is,t} / p_{s,t}^Y$. We use the nominal wage bill rather than employment to correct for differences in workforce quality across firms, as argued in, among others, [Hsieh and Klenow \(2009\)](#) and [Gopinath et al. \(2017\)](#). Finally, we measure firm-specific real capital stock $K_{is,t}$ using the book value of tangible fixed assets, deflated by the aggregate deflator for gross fixed capital formation of non-financial corporations, p_t^K . This implies that we approximate $K_{is,t}$ as $p_{is,t}^K K_{is,t} / p_t^K$.

Each firm behaves as a monopolist in its differentiated product and is subject to two types of distortions: a capital wedge $\tau_{is,t}^K$, which alters the relative marginal revenue product of capital versus labor, and an output wedge $\tau_{is,t}^Y$, which distorts the marginal products of capital and labor proportionally. Market distortions enter the firm’s profit function as:

$$\pi_{is,t} = (1 - \tau_{is,t}^Y) p_{is,t}^Y Y_{is,t} - w_{s,t} L_{is,t} - (1 + \tau_{is,t}^K) r_{s,t} K_{is,t}. \quad (18)$$

The first-order conditions for profit maximization with respect to labor and capital are:

$$MRPL_{is,t} = p_{is,t}^Y \frac{\partial Y}{\partial L} = (1 - \alpha_s) \left(\frac{\sigma - 1}{\sigma} \right) \left(\frac{p_{is,t}^Y Y_{is,t}}{L_{is,t}} \right) = \left(\frac{1}{1 - \tau_{is,t}^Y} \right) w_{s,t}, \quad (19)$$

$$MRPK_{is,t} = p_{is,t}^Y \frac{\partial Y}{\partial K} = \alpha_s \left(\frac{\sigma - 1}{\sigma} \right) \left(\frac{p_{is,t}^Y Y_{is,t}}{K_{is,t}} \right) = \left(\frac{1 + \tau_{is,t}^K}{1 - \tau_{is,t}^Y} \right) r_{s,t}. \quad (20)$$

We estimate $MRPL$ and $MRPK$ using firm-level data on value added, labor, and capital. In addition, we compute α_s as one minus the share of wage costs in value added at the five-digit industry level. The elasticity of substitution between varieties is assumed to be the same for all firms, sectors, and years and is set equal to $\sigma = 3$.

¹Industry deflators are publicly available data from Statistics Netherlands.

Following [Foster et al. \(2008\)](#), the Hsieh–Klenow model distinguishes between revenue productivity (TFPR) and physical productivity (TFPQ). TFPQ can be defined by rewriting equation (17):

$$TFPQ_{is,t} = \omega_{is,t} = \kappa_{s,t} \frac{(p_{is,t}^Y Y_{is,t})^{\frac{\sigma}{\sigma-1}}}{K_{is,t}^{\alpha_s} L_{is,t}^{1-\alpha_s}}, \quad (21)$$

where $\kappa_{s,t} = (p_{s,t}^Y Y_{s,t})^{\frac{1}{\sigma-1}} / p_{s,t}^Y$ is an industry-specific scaling factor. Because firm-level prices are not directly observed, we do not directly observe real output at the firm level, only nominal output $p_{is,t}^Y Y_{is,t}$. Assuming iso-elastic inverse demand,² we therefore raise $p_{is,t}^Y Y_{is,t}$ to the power $\sigma/(\sigma-1)$ to obtain a measure of physical output.

TFPR is then defined as:

$$TFPR_{is,t} = p_{is,t}^Y \omega_{is,t} = \left(\frac{\sigma}{\sigma-1} \right) \left(\frac{MRPK_{is,t}}{\alpha_s} \right)^{\alpha_s} \left(\frac{MRPL_{is,t}}{1-\alpha_s} \right)^{1-\alpha_s}. \quad (22)$$

In the model of [Hsieh en Klenow \(2009\)](#), TFPR does not vary across firms within an industry unless firms face output and/or capital distortions.³

Within industries, large differences in the marginal revenue product of capital can arise when capital remains stuck for prolonged periods in low-productivity firms. Under efficient allocation, these firms would shrink, releasing capital to more productive firms, until marginal revenue products of capital are equalized within the sector. A similar intuition applies to labor. Recent studies link wedges in the labor first-order condition to labor market power, see for example [Dobbelaere en Kiyota \(2018\)](#) and [Mertens \(2020\)](#). Empirically, we indeed observe both negative and positive labor wedges.

Following [Hsieh en Klenow \(2009\)](#), we estimate the impact of misallocation on the level of TFPQ by defining the “efficient” level of TFPQ as the level that would prevail under an allocation with no dispersion in $MRPK$, $MRPL$, and $TFPR$, such that $TFPR_{is,t} = \overline{TFPR}_{s,t}$, where:

$$\overline{TFPR}_{s,t} = \frac{p_{s,t}^Y Y_{s,t}}{K_{s,t}^{\alpha_s} L_{s,t}^{1-\alpha_s}}. \quad (23)$$

It can then be shown that the resulting difference in $\log(TFPQ)$ —the misallocation gain $\Lambda_{s,t}$ —can be expressed as a function of the variables introduced in equations (17)–(23). See, for example, equations (1)–(11) in [Gopinath et al. \(2017\)](#) for a detailed derivation.

²[Haltiwanger et al. \(2018\)](#) provide a critical assessment of this assumption and show that part of the observed dispersion in TFPR can be attributed to demand shocks.

³Zero dispersion in $MRPK$, $MRPL$, and $TFPR$ only requires that the wedges be identical across firms within a sector; a distortion may still exist if it affects all firms equally.

Note that equation (23) accounts only for misallocation within industries. However, [Calligaris et al. \(2018\)](#) show that this within-industry component is typically large. Applying their TFPR variance decomposition (see equation (2) in [Calligaris et al. \(2018\)](#)), we find a very similar result for the Netherlands: on average over the sample period, 80% of TFPR dispersion is attributable to the within component.

Our misallocation measure represents an upper bound on within-industry misallocation, since all variation in marginal revenue products is attributed to misallocation. This is a strong assumption. The assumptions underlying the [Hsieh en Klenow \(2009\)](#) model are quite stringent: all firms within a sector share the same markup and capital intensity, there are no adjustment costs, and the elasticity of substitution between capital and labor follows from a Cobb–Douglas production function with constant returns to scale. Moreover, input composition effects are abstracted from. For example, differences in the mix of high- and low-skilled labor with different marginal productivities can generate dispersion in $MRPL$, even when allocation is in fact efficient. To the extent that such composition effects are constant over time, changes in $MRPL$ can still be interpreted as changes in misallocation. Despite these limitations, we consider the model sufficiently suitable for mapping misallocation in the economy.

B Regression Analysis: Dependent Variable Markups

In this section, we describe the regression analysis underlying the finding that an increase in domestic market concentration does not necessarily have negative effects, and that in internationally competitive industries higher concentration is often associated with selection and scale effects without translating into higher markups (see the Summary and Chapter 2 of the DNB Analysis).

Table I presents the results of our fixed-effects regression analysis, in which we relate sector-year (SBI-3D) markups to the entry rate (entries), the exit rate (exits), and the markdown.

The baseline regression results are reported in column (1) of Table I. This specification includes both year and sector fixed effects. The (adjusted) R^2 of this regression is approximately 0.86.⁴ For comparison, column (4) shows results from a regression that includes only year fixed effects. In this specification, a large share of the explanatory power disappears and the (adjusted) R^2 drops to about 0.20. This underscores the importance of including industry dummies and, more generally, controlling for unobserved heterogeneity across sectors.

Regression (1) shows that both higher market concentration (HHI) and a higher markdown, conditional on year and sector fixed effects, have a statistically significant negative effect on the markup. These results are highly robust: the estimated effects remain statistically significant across all specifications in columns (1) through (6), both in models with year and sector fixed effects and in models with sector fixed effects only.

In regressions (2) and (3), we examine whether the effect of market concentration on markups differs across industries with varying degrees of domestic orientation. To do so, we use the taxonomy developed by [Calligaris et al. \(2024\)](#), which determines for each industry the share of sales on the domestic, European, and global markets. Based on this taxonomy, we define two variables. The variable *Shielded industry (dummy)* equals one if the domestic sales share exceeds the median value in our sample (0.81), and zero otherwise. The variable *Shielded industry (share)* is the continuous counterpart and captures the share of sales on the domestic market.

Columns (2) and (3) report the results of regressions in which, respectively, the dummy and the continuous measure of domestic orientation are interacted with market concentration. The results show that an increase in market concentration has a negative effect on markups, but that this effect depends on the degree of international orientation of the in-

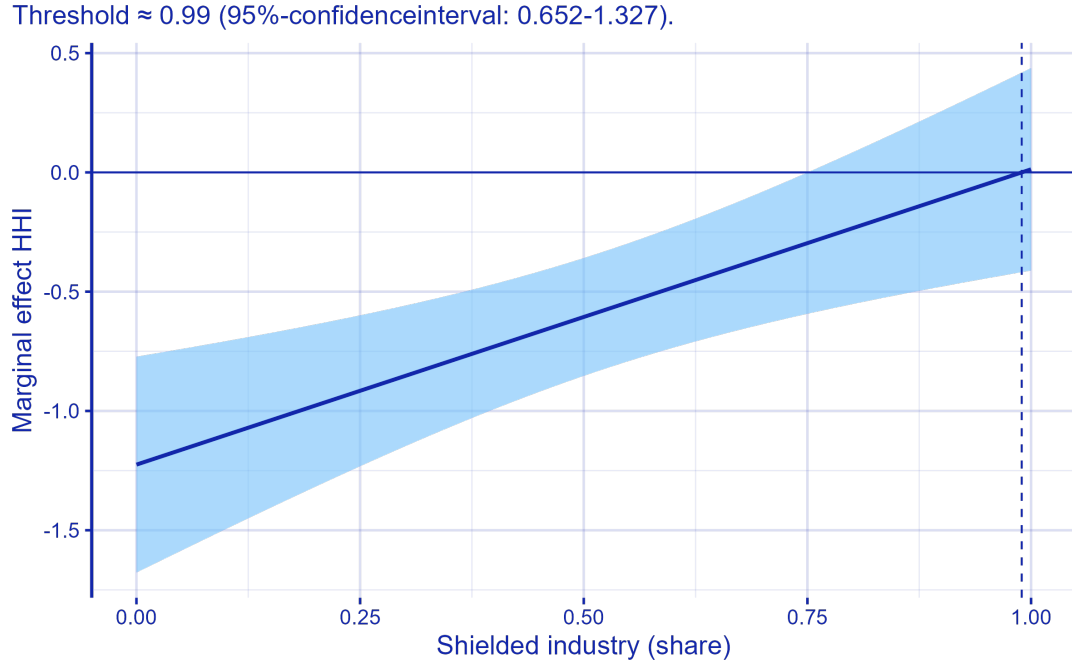
⁴Minor differences are due to rounding in the table.

Tabel I: Regression results: markup

	(1)	(2)	(3)	(4)	(5)	(6)
Market concentration (HHI)	-0.454** (0.169)	-0.792*** (0.185)	-0.425** (0.134)	-0.559** (0.199)	-0.535* (0.242)	-0.558** (0.195)
Births	-0.233* (0.107)	-0.219* (0.107)	-0.230* (0.107)	0.467* (0.218)	0.477* (0.216)	0.504* (0.215)
Deaths	-0.177 (0.208)	-0.143 (0.206)	-0.156 (0.204)	0.719 (0.435)	0.728 (0.442)	0.669 (0.445)
Markdown	-0.034** (0.011)	-0.034** (0.011)	-0.033** (0.011)	-0.051*** (0.012)	-0.052*** (0.012)	-0.053*** (0.012)
Shielded sector (dummy)					0.000 (0.028)	
Shielded sector (share)						0.037 (0.039)
Marketconc. (HHI) $\times$ Shielded sector (dummy)		0.871** (0.281)			-0.076 (0.408)	
Markteconc. (HHI) $\times$ Shielded sector (sector)			1.238** (0.370)			-0.582 (0.785)
Num.Obs.	1789	1789	1789	1795	1795	1795
R2	0.816	0.819	0.819	0.208	0.208	0.212
R2 Adj.	0.801	0.804	0.803	0.200	0.199	0.203
R2 Within	0.062	0.078	0.074	0.204	0.204	0.208
R2 Within Adj.	0.059	0.075	0.071	0.202	0.201	0.205
AIC	-4394.3	-4423.3	-4416.4	-2004.2	-2000.8	-2010.0
BIC	-3614.8	-3638.3	-3631.4	-1894.3	-1880.0	-1889.2
RMSE	0.07	0.06	0.07	0.14	0.14	0.14
Std.Errors	by: sector	by: sector	by: sector	by: sector	by: sector	by: sector
FE: year	X	X	X	X	X	X
FE: sector	X	X	X			

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Figuur 1: Interaction effect Market Concentration (HHI) $\times$ Shielded industry (share)



dustry. The interaction terms *Market concentration (HHI) $\times$ Shielded industry (dummy)* and *Market concentration (HHI) $\times$ Shielded industry (share)* are positive and statistically significant, indicating that the negative effect of market concentration on markups is weaker—and for very strongly domestically oriented industries even reverses sign.

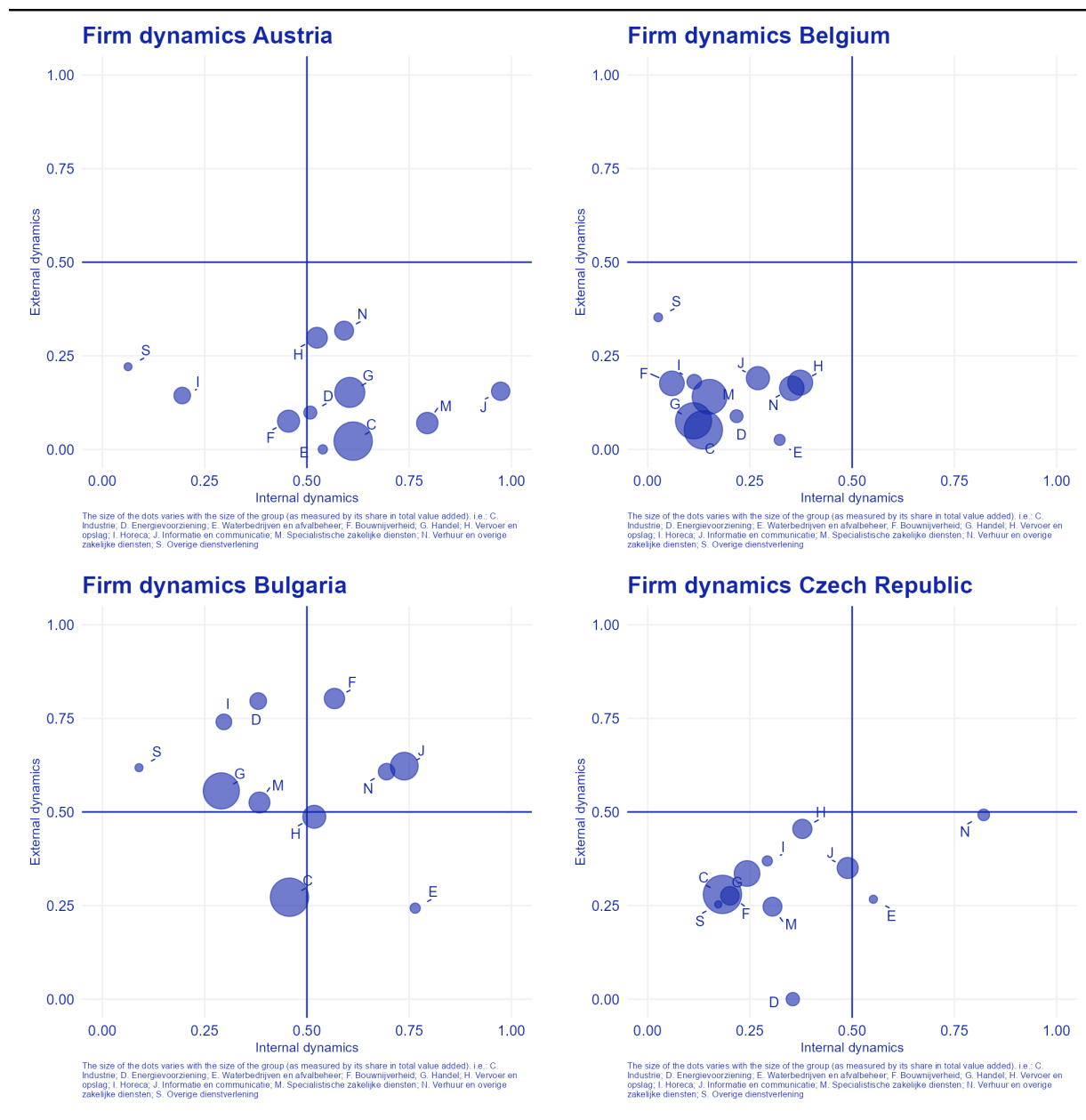
Figure 1 illustrates how these results can be interpreted. The figure provides a graphical representation of regression (3) and shows the marginal effect of market concentration on markups for different levels of domestic orientation. The point estimate suggests that the effect becomes positive at a domestic orientation of approximately 0.99 (indicated by the vertical dashed line). This implies that only for industries that are almost entirely (around 99%) oriented toward the domestic market does an increase in market concentration coincide with higher markups. For industries with lower domestic orientation, higher market concentration does not lead to higher markups.

Figure 1 also shows the uncertainty surrounding these estimates. The upper bound of the 95% confidence interval crosses the zero line at a domestic orientation of around 0.75. Taking this uncertainty into account, it is possible that for industries with more than roughly 75% domestic orientation market concentration has a positive effect on markups. At the same time, the effect may remain negative across all levels of domestic orientation, since the lower bound of the confidence interval lies below zero for all values.

C Extra Figures

This chapter contains the additional figures referred to in the main text of the DNB Analysis.

Figure II: Internal en external dynamics dynamics per country



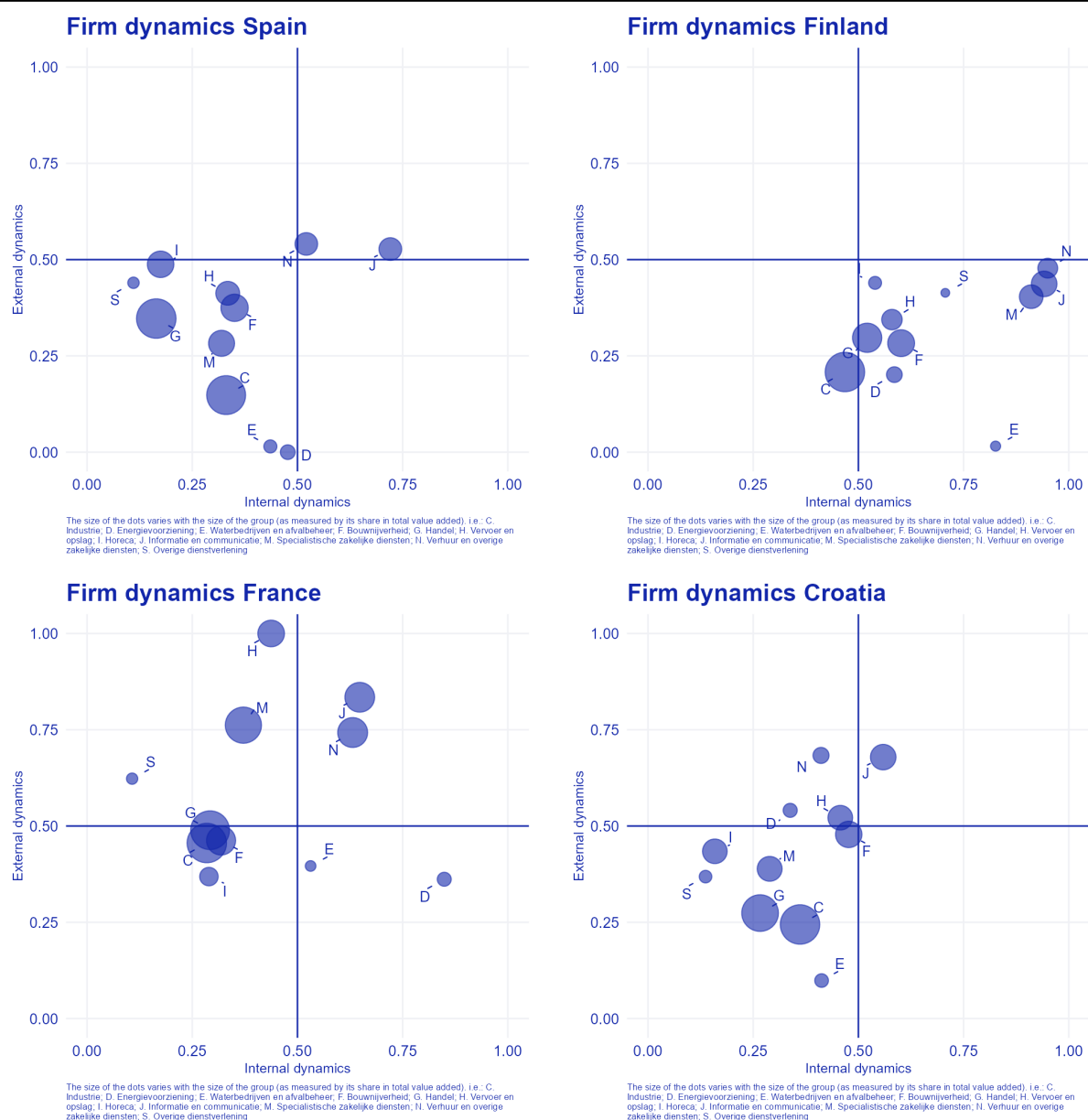
Figuer continued on next page

Figure II: Internal en external dynamics dynamics per country (continued)



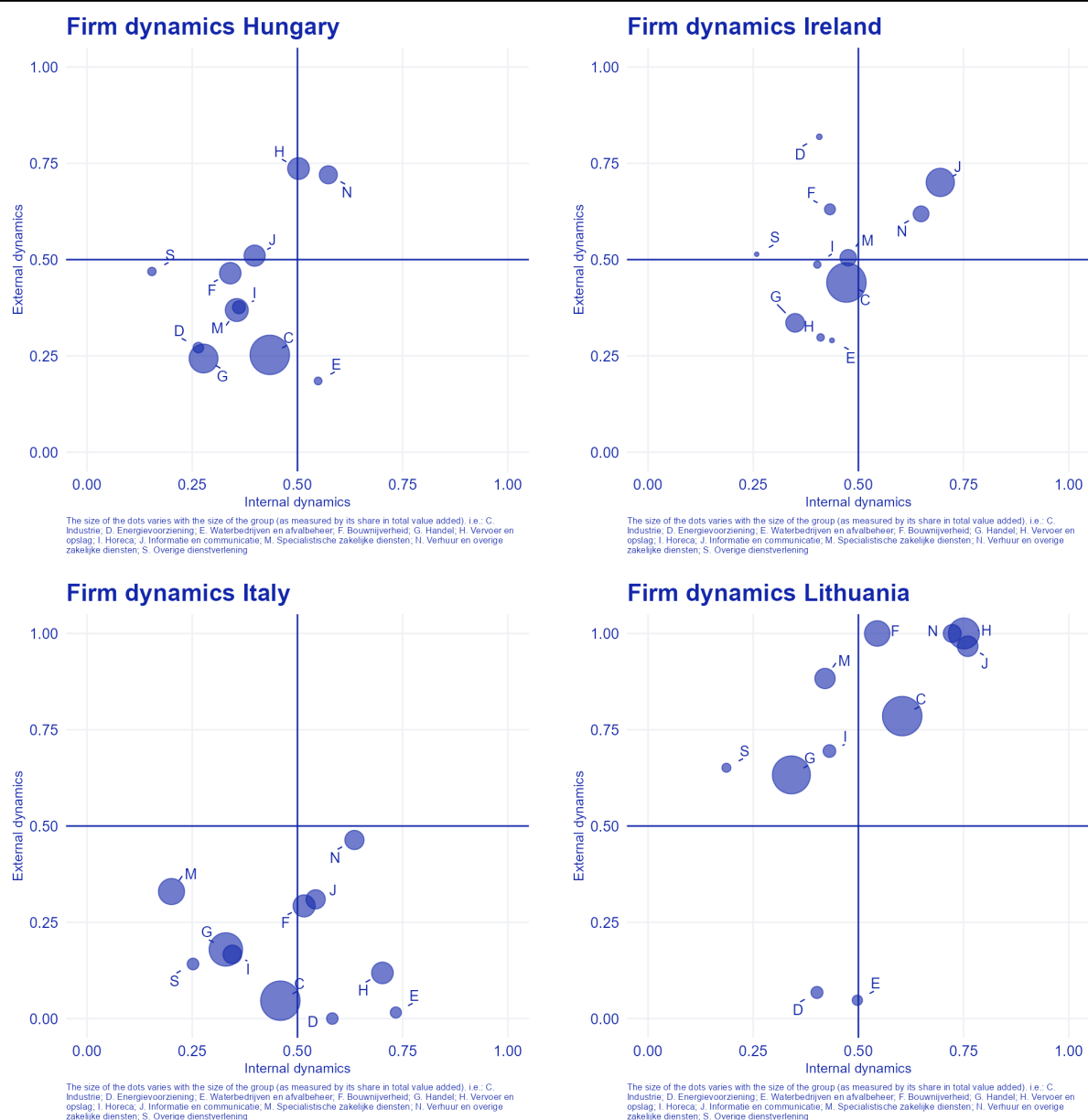
Figuer continued on next page

Figure II: Internal en external dynamics dynamics per country (continued)



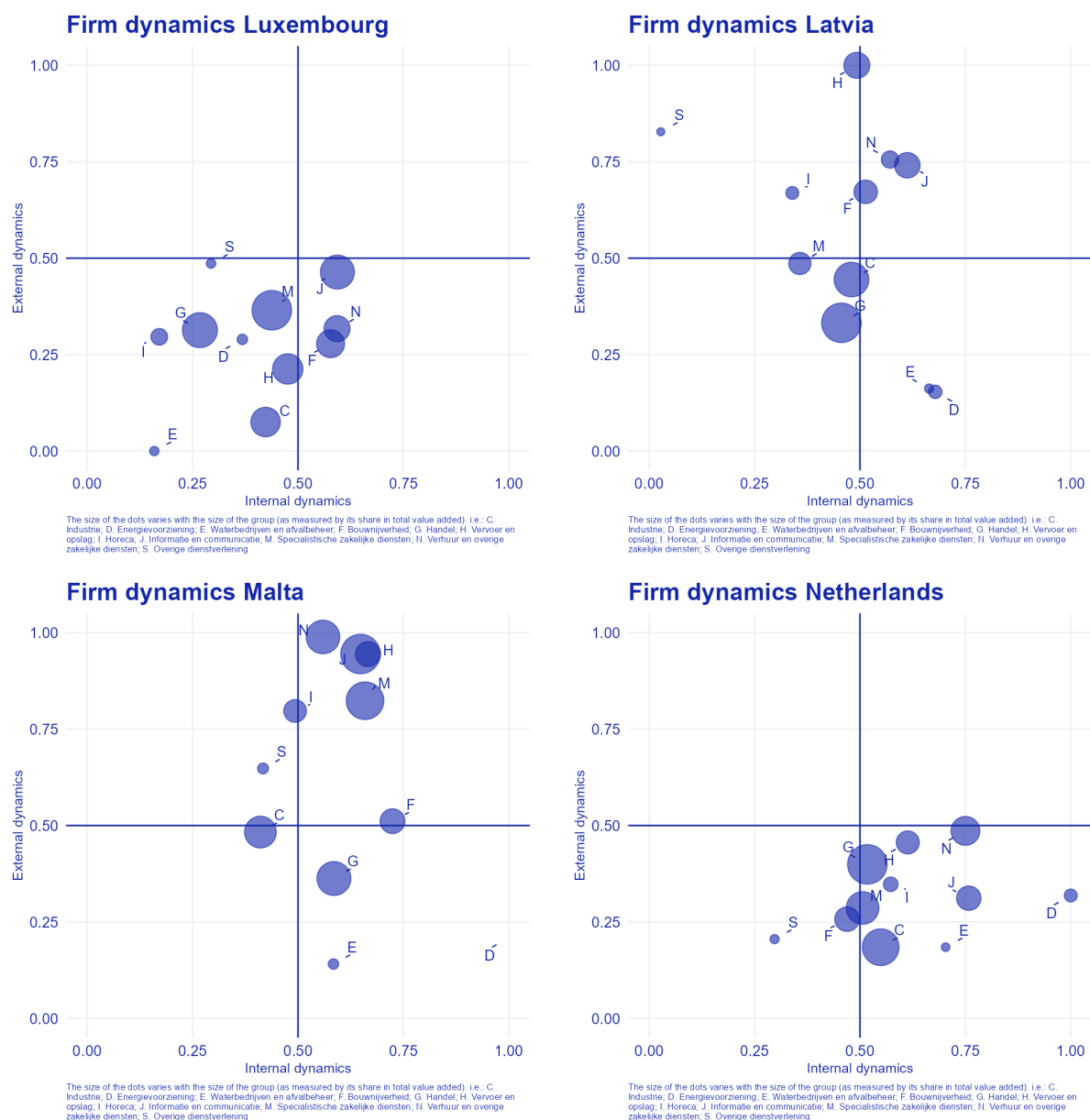
Figuer continued on next page

Figure II: Internal en external dynamics dynamics per country (continued)



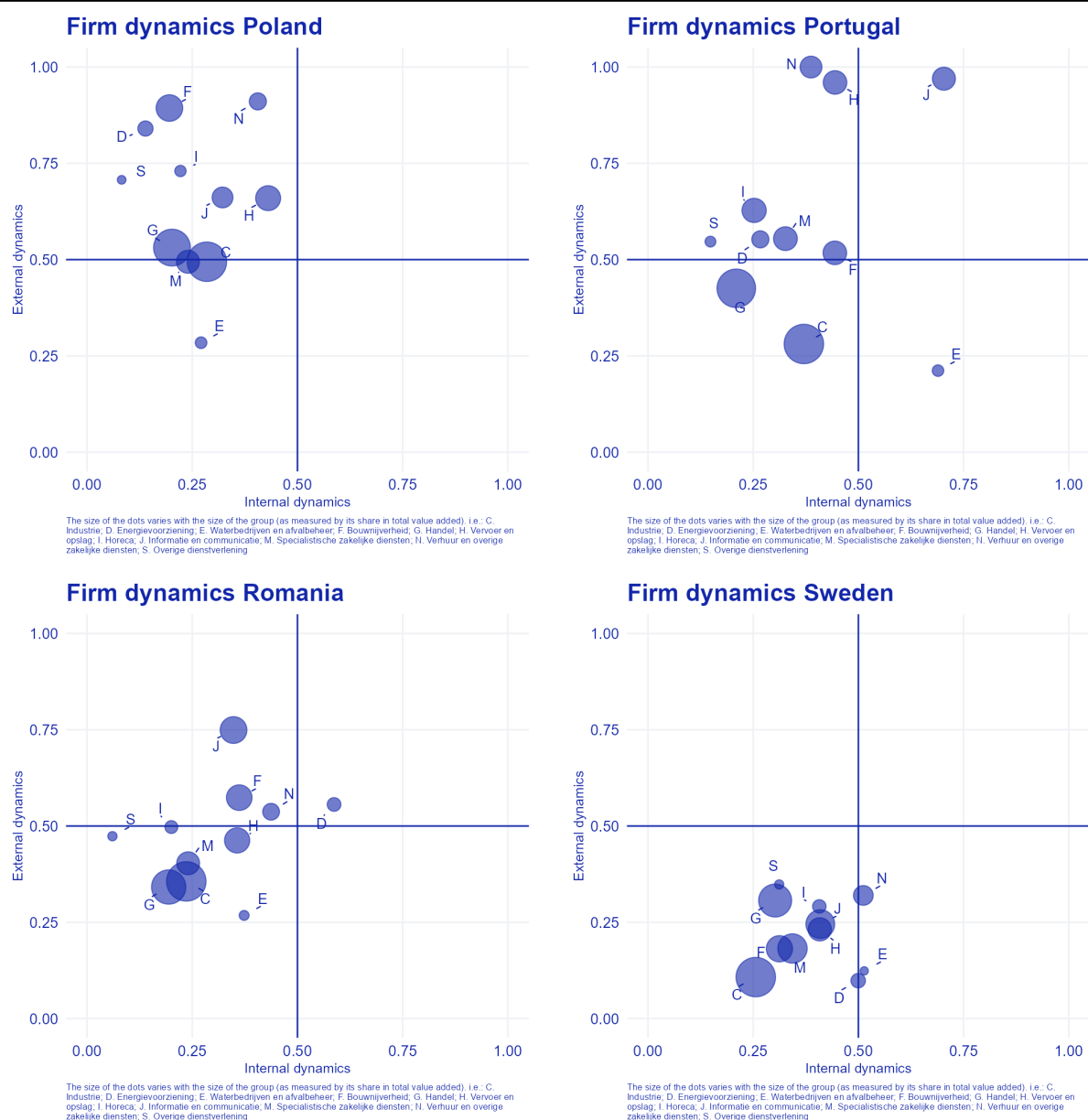
Figuer continued on next page

Figure II: Internal en external dynamics dynamics per country (continued)



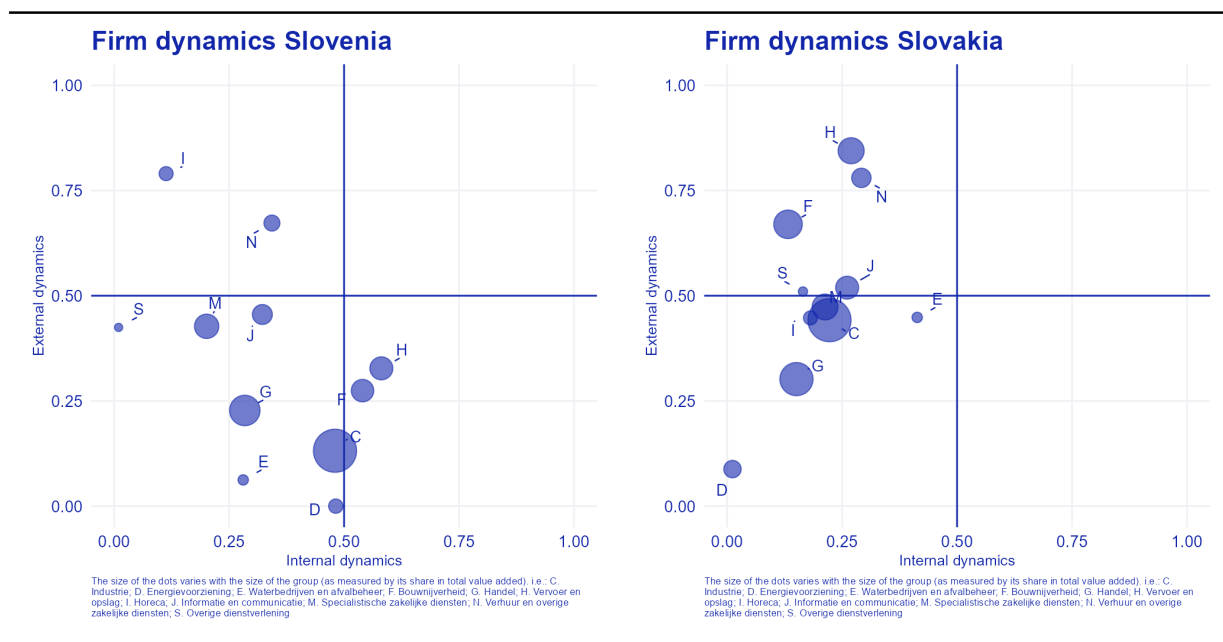
Figuer continued on next page

Figure II: Internal en external dynamics dynamics per country (continued)

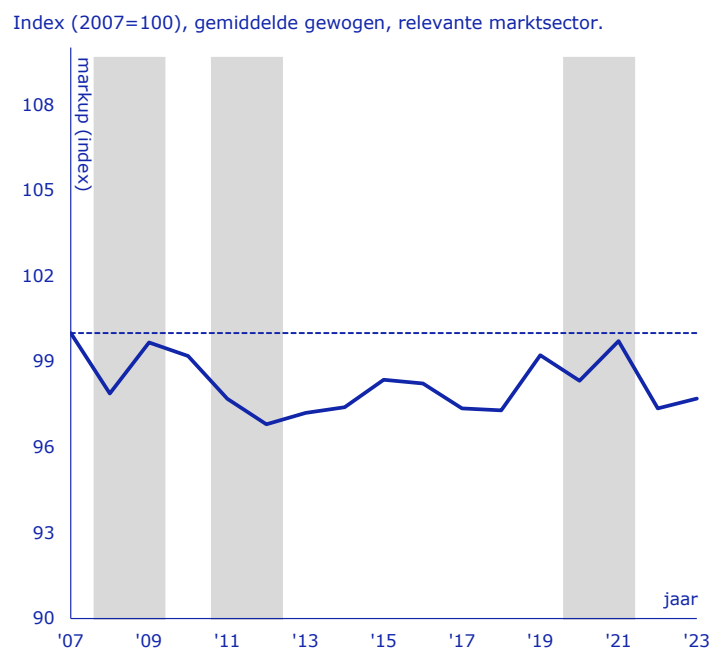


Figuer continued on next page

Figure II: Internal en external dynamics dynamics per country (continued)



Figuur 3: Markups in the Netherlands, method [De Loecker et al. \(2026\)](#)

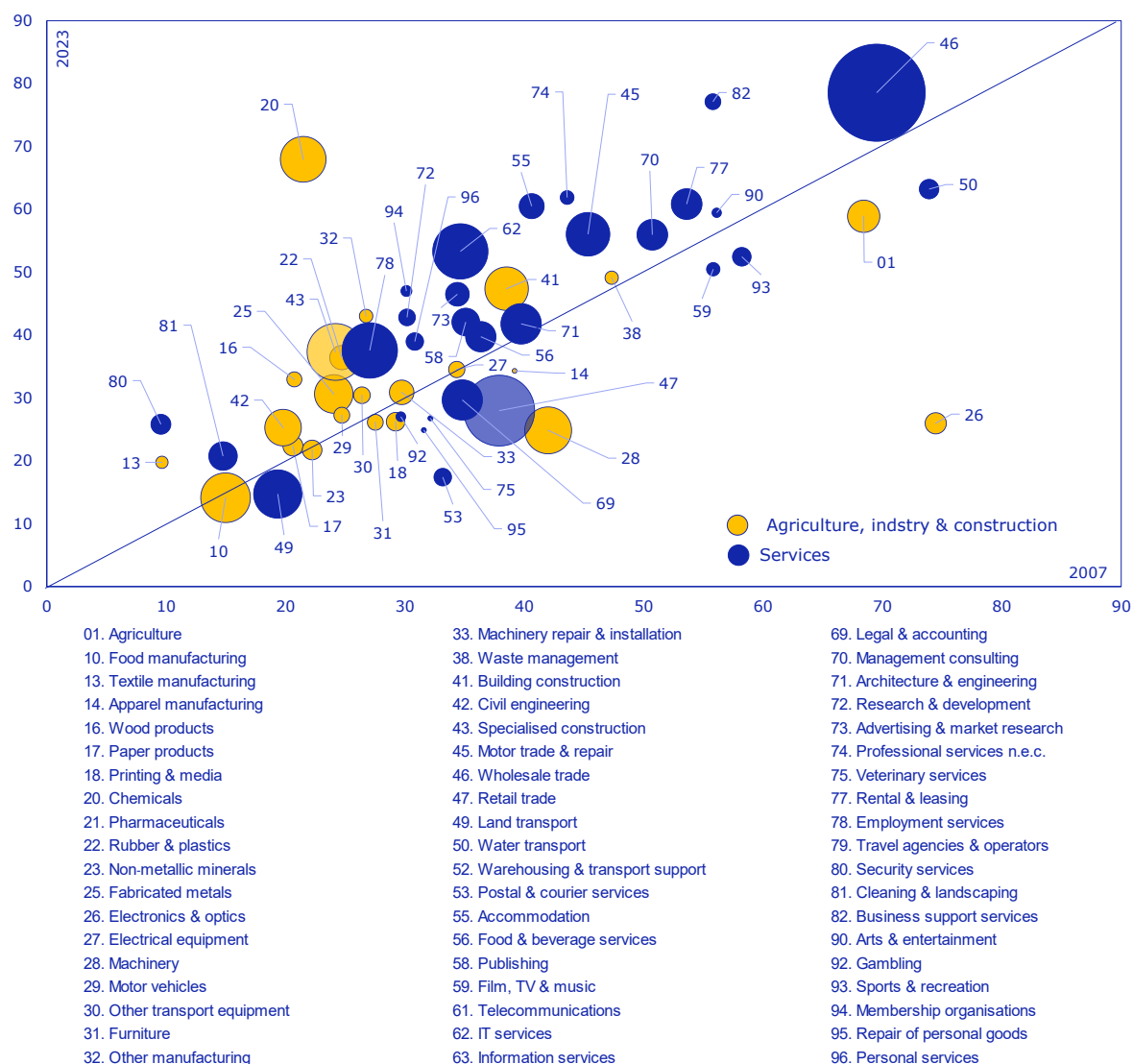


Bron: CBS-microdata, berekeningen DNB.

Toelichting: Gearceerde gebieden duiden op recessies: de financiële crisis (2008–2009), EU-schuldencrisis (2011–2012), en de COVID-pandemie (2020–2021). De blauwe horizontale stippellijn kruist de y-as bij 100. De blauwe lijn geeft de gemiddelde gewogen markup weer. De markup is berekend volgens de methode van De Loecker et al. (2026). Zie Paragraaf A.2.2 voor meer details

Figuur 4: Misallocation per industry, NACE-2D level, 2003 versus 2007

In % total factorproductivty (tfp), method Hsieh en Klenow (2009)

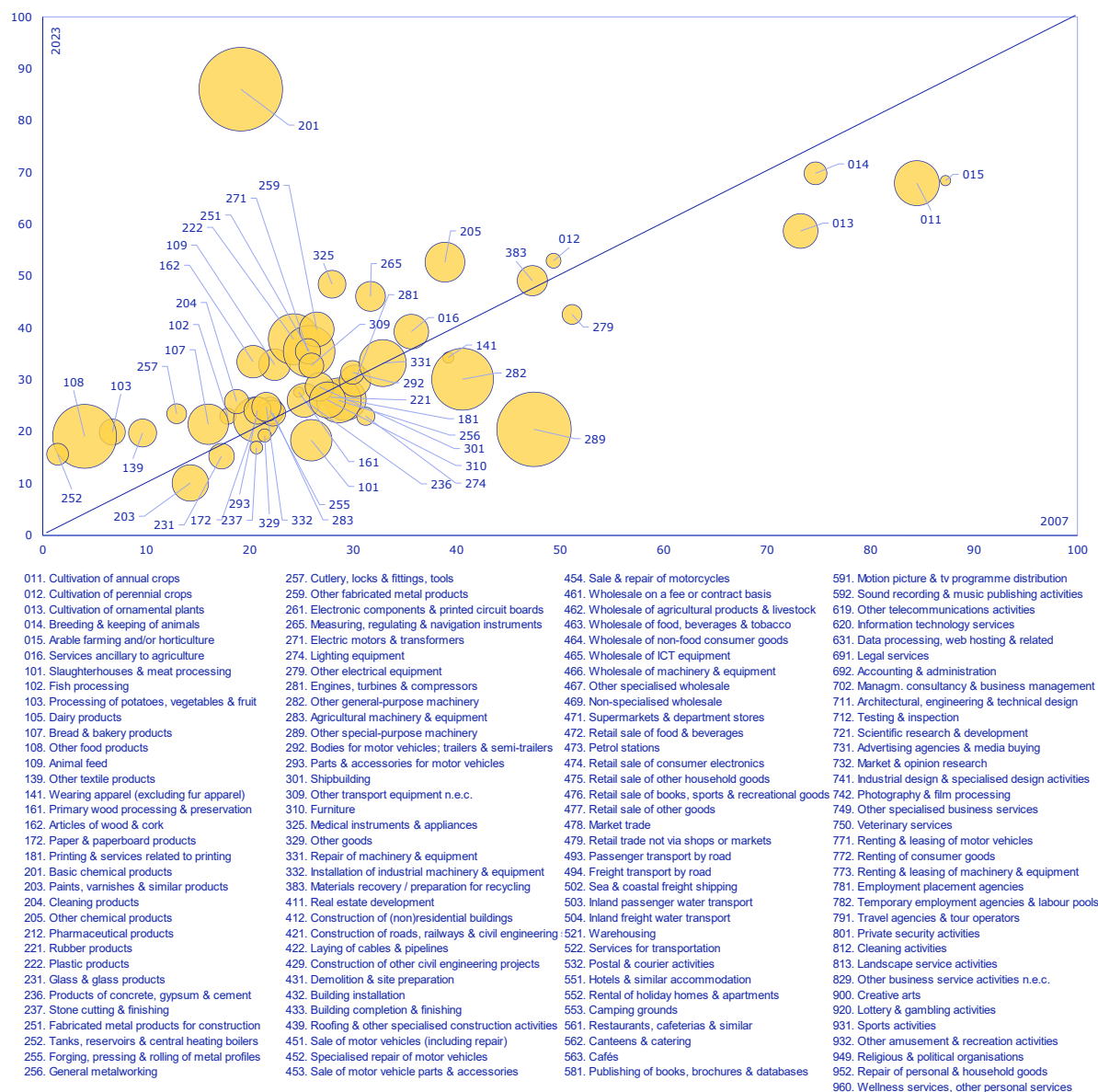


Source: Microdata Statistics Netherlands, DNB calculations.

Note: Misallocation in percent gain/loss of tfp against efficient allocation. The size of the circles indicates the industry's relative share in value added in the Netherlands.

Figuur 5: Misallocation per industry, NACE-3D level, agriculture, industry and construction, 2003 versus 2007

In % total factorproductivty (tfp), method Hsieh en Klenow (2009)

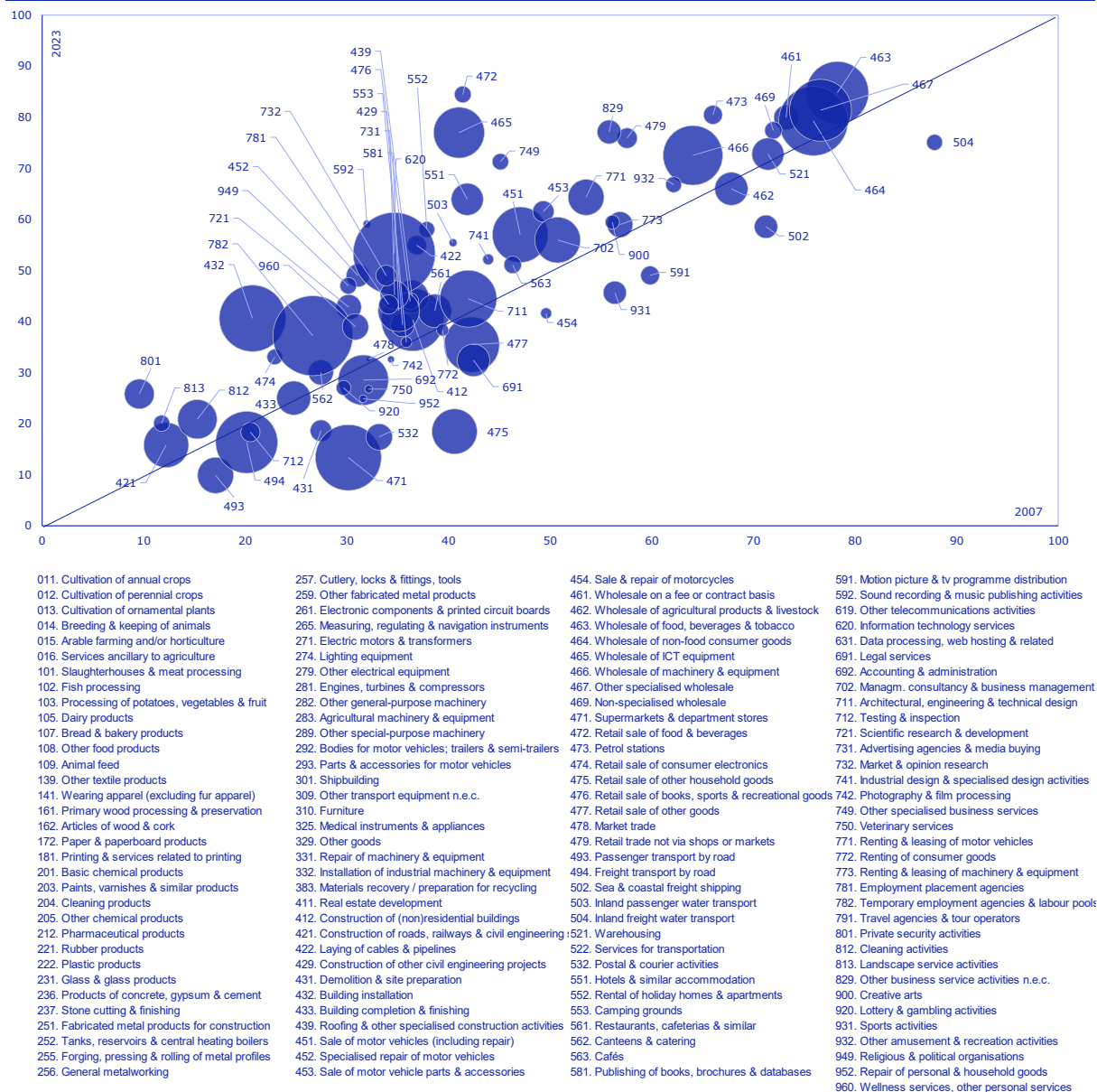


Source: Microdata Statistics Netherlands, DNB calculations.

Note: Misallocation in percent gain/loss of tfp against efficient allocation. The size of the circles indicates the industry's relative share in value added in the Netherlands.

Figuur 6: Misallocation per industry, NACE-3D level, services sector, 2003 versus 2007

In % total factorproductivty (tfp), method Hsieh en Klenow (2009)



Source: Microdata Statistics Netherlands, DNB calculations.

Note: Misallocation in percent gain/loss of tfp against efficient allocation. The size of the circles indicates the industry's relative share in value added in the Netherlands.

Bibliografie

- Akerberg, D. A., K. Caves, en G. Frazer (2015). Identification properties of recent production function estimators. *Econometrica* 83(6), 2411–2451.
- Adema, Y., L. Bettendorf, E. van Bezooijen, D. Freeman, en B. Wache (2015). Dynamics, productivity, and innovation in the Dutch economy. Discussion paper, Centraal Planbureau.
- Calligaris, S. (2015). Misallocation and total factor productivity in Italy: Evidence from firm-level data. *Labour* 29(4), 367–393.
- Calligaris, S., C. Criscuolo, J. De Lyon, A. Greppi, en O. Pallanch (2024). Defining the geographical level of competition: A taxonomy of industries. Science, Technology and Industry Working Papers 5, Organisation for Economic Co-operation and Development.
- Calligaris, S., M. Del Gatto, F. Hassan, G. I. Ottaviano, en F. Schivardi (2018). The productivity puzzle and misallocation: An Italian perspective. *Economic Policy* 33(96), 635–684.
- De Loecker, J., J. Eeckhout, en G. Unger (2020). The rise of market power and the macroeconomic implications. *The Quarterly Journal of Economics* 135(2), 561–644.
- De Loecker, J., C. Fuss, N. Quiller-Doust, en L. Treuren (2026). The anatomy of costs and firm performance: Evidence from Belgium. *International Journal of Industrial Organization* 105.
- De Loecker, J. en F. Warzynski (2012). Markups and firm-level export status. *American Economic Review* 102(6), 2437–2471.
- Dias, D. A., C. R. Marques, en C. Richmond (2016). Misallocation and productivity in the lead up to the eurozone crisis. *Journal of Macroeconomics* 49, 46–70.
- Dobbelaere, S. en K. Kiyota (2018). Labor market imperfections, markups and productivity in multinationals and exporters. *Labour Economics* 53, 198–212.
- Foster, L., J. Haltiwanger, en C. Syverson (2008). Reallocation, firm turnover, and efficiency: Selection of productivity or profitability? *American Economic Review* 98(1), 394–425.
- Gamberoni, E., C. Giordano, en P. Lopez-Garcia (2016). Capital and labour (mis)allocation in the euro area: Some stylized facts and determinants. Working Paper Series 1981, European Central Bank.
- Gopinath, G., S. Kalemli Özcan, L. Karabarbounis, en C. Villegas-Sanchez (2017). Capital allocation and productivity in South Europe. *The Quarterly Journal of Economics* 132(4), 1915–1967.
- Gorodnichenko, Y., D. Revoltella, J. Svejnar, en C. T. Weiss (2018). Resource misallocation in European firms: The role of constraints, firm characteristics and managerial decisions. NBER Working Papers 24444, National Bureau of Economic Research, Inc.

- Haltiwanger, J., R. Kulick, en C. Syverson (2018). Misallocation measures: The distortion that ate the residual. NBER Working Papers 24199, National Bureau of Economic Research, Inc.
- Hsieh, C. en P. J. Klenow (2009). Misallocation and manufacturing TFP in China and India. *The Quarterly Journal of Economics* 124(4), 1403–1448.
- Mertens, M. (2020). Micro-mechanisms behind declining labor shares: Rising market power and changing modes of production. Working paper, Halle Institute for Economic Research.
- Olley, G. S. en A. Pakes (1996). The dynamics of productivity in the telecommunications equipment industry. *Econometrica* 64(6), 1263–1279.
- van Heuvelen, G. H., L. Bettendorf, en G. Meijerink (2019). Estimating markups in the Netherlands. CPB Background Document March 2019, CPB Netherlands Bureau for Economic Policy Analysis.
- Yeh, C., C. Macaluso, en B. Hershbein (2022). Monopsony in the US labor market. *American Economic Review* 112(7), 2099–2138.